PEARSON EDEXCEL FUNCTIONAL SKILLS MATHEMATICS
MARK SCHEME – LEVEL 1 SET 5

Marking Guidance for Functional Skills Mathematics Level 1

General

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded; exemplification will not be exhaustive. When examiners are in doubt regarding the application of the mark scheme, the response should be escalated to a senior examiner to review.
- Mark schemes should be applied positively. Learners must be rewarded for what they have shown they can do rather than penalised for omissions.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the learner's response is not worthy of credit according to the mark scheme. If there is a wrong answer (or no answer) indicated in the answer box, always check the working in the body of the script (and on any diagrams), and award any marks appropriate from the mark scheme.
- Working is always expected. For short questions, where working may not be seen, correct answers may still be awarded full marks. For longer questions, an answer in brackets from the mark scheme seen in the body of the working, implies a correct process and the appropriate marks may be awarded.
- **Questions that specifically state that working is required**: learners who do not show working will get no marks – full details will be given in the mark scheme for each individual question.

Applying the Mark Scheme

- The mark scheme has a column for **Process** and a column for **Evidence**. In most questions the majority of marks are awarded for the process the learner uses to reach an answer. The evidence column shows the *most likely* examples that will be seen. If the learner gives different evidence valid for the process, examiners should award the mark(s).
- If working **is crossed out and still legible**, then it should be marked, as long as it has not been replaced by alternative work.
- If there is a **choice of methods** shown, then mark the work leading to the answer given in the answer box or working box. If there is no definitive answer then marks should be awarded for the lowest scoring method shown.
- A suspected **misread**, e.g. 528 instead of 523, may still gain process marks provided the question has not been simplified. Examiners should send any instance of a suspected misread to a senior examiner to review.
- It may be appropriate to **ignore subsequent work (isw)** when the learner's additional work does not change the meaning of their answer.

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- **Correct** working followed by an **incorrect decision** may be seen, showing that the learner can calculate but does not understand the functional demand of the question. The mark scheme will make clear how to mark these questions.
- **Transcription** errors occur when the learner presents a correct answer in working, and writes it incorrectly on the answer box e.g. 698 in the body and 689 in the answer box; mark the better answer if clearly only a transcription error. Examiners should send any instance of transcriptions errors to a senior examiner to review.
- **Incorrect method** if it is clear from the working that the correct answer has been obtained from incorrect working, award 0 marks. Examiners must escalate the response to a senior examiner to review.
- **Follow through marks (ft)** must only be awarded when explicitly allowed in the mark scheme. Where the process uses the learner's answer from a previous step, this is clearly shown.
  - Speech marks are used to show that previously incorrect numerical work is being followed through, for example '240' means their 240 coming from a correct or set of correct processes.
  - When words are used in { } then this value does not need to come from a correct process but should be the value the learner believes to be required. The constraints on this value will be detailed in the mark scheme. For example, {volume} means the figure may not come from a correct process but is clearly the value learners believe should be used as the volume.
- Marks can usually be awarded where units are not shown. Where units are required this will be stated. For example, 5(m) indicates that the units do not have to be stated for the mark to be awarded.
- Learners may present their answers or working in many **equivalent** ways. This is denoted oe in the mark scheme. Repeated addition for multiplication and repeated subtraction for division are common alternative approaches. The mark scheme will specify the minimum required to award these marks.
- A **range** of answers is often allowed, when a range of answers is given e.g. [12.5, 13] this is the inclusive closed interval.
- **Accuracy** of figures. Accept an answer which has been rounded or truncated from the correct figure unless other guidance is given. For example, for 12.66.. accept 12.6, 12.7, 12.66, 12.67 or any other more accurate figure.
- **Probability** answers must be given as a fraction, percentage or decimal. If a learner gives a decimal equivalent to a probability, this should be written to at least 2 decimal places (unless tenths). If a learner gives the answer as a percentage a % must be used. Incorrect notation should lose the accuracy marks, but be awarded any implied process marks. If a probability fraction is given then cancelled incorrectly, ignore the incorrectly cancelled answer.
- **Graphs.** A linear scale must be linear in the range where data is plotted, and use consistent intervals. The scale may not start at 0 and not all intervals must be labelled. The minimum requirements will be given, but examiners should give credit if a title is given which makes the label obvious.

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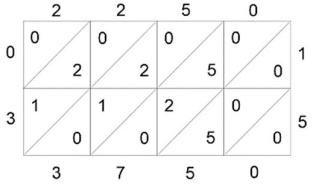
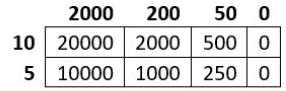
Section A (Non-Calculator)

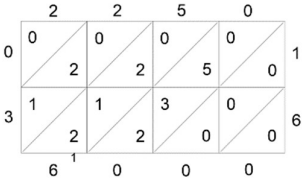
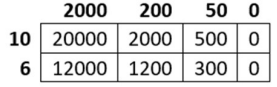
PMAT1/N05				
Question	**Process**	**Mark**	**Mark Grid**	**Evidence**
Q1(a)	All figures in correct order	1	A	1.7(00), 1.703, 1.76(0), 1.878
Q1(b)	At least 3 correct conversions into common denominator, decimals or percentages or 3 of the original fractions in the correct order or 2 of the original fractions in correct order and the other 2 written in a common format	1 or	B	e.g. 3 of $\frac{10}{20}$, $\frac{8}{20}$, $\frac{6}{20}$, $\frac{15}{20}$ oe **or** 0.5, 0.4, 0.3, 0.75 **or** 50, 40, 30, 75 **OR** e.g. $\frac{3}{10}$, $\frac{4}{10}$ seen in either order **and** $\frac{1}{2}$, $\frac{3}{4}$
	All figures in correct order	2	BC	$\frac{3}{10}$, $\frac{2}{5}$, $\frac{1}{2}$, $\frac{3}{4}$ **OR** 0.3, 0.4, 0.5, 0.75 **OR** 30(%), 40(%), 50(%), 75(%) Allow figures to be written in any equivalent or mixed format to correct answer
	Total marks for question	**3**		

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Question	Process	Mark	Mark Grid	Evidence
Q2(a)	Accurate figure	1	A	98
Q2(b)	Begins process to find range	1 or	B	−13 − −3 (= −10) **OR** −3 − −13 (=10) **OR** −3 to −13
	Accurate figure	2	BC	10
	Total marks for question	**3**		

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<table>
<tr><th>Question</th><th>Process</th><th>Mark</th><th>Mark Grid</th><th>Evidence</th></tr>
<tr><td rowspan="4">Q3</td><td>Starts to work with perimeter</td><td>1</td><td>A</td><td>e.g. 2 × 8 (=16)</td></tr>
<tr><td>Begins to work out cost of fencing</td><td>1</td><td>B</td><td>e.g. 2 × 22.50 (= 45) OR
8 × 22.50 (= 180) OR
‘15’ × 22.50 OR
‘16’ × 22.50</td></tr>
<tr><td>Complete method with relative place value correct including addition of all the appropriate elements of the calculation e.g. two lines of 1st method, internal numbers of grids, or complete structure shown of partitioning methods, to multiply 15 × 22.50 or 16 × 22.50 (ignoring decimal point)</td><td>2 or</td><td>BC</td><td>11250
<u>22500</u>
33750
 
 
20000 + 2000 + 500 +
10000 + 1000 + 250 = 33750

13500
<u>22500</u>
36000
 
 
20000 + 2000 + 500 +
12000 + 1200 + 300 = 36000</td></tr>
<tr><td>Accurate figure including correct placement of decimal point</td><td>3</td><td>BCD</td><td>337.50</td></tr>
<tr><td colspan="3">Total marks for question</td><td colspan="2">4</td></tr>
</table>

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Question	Process	Mark	Mark Grid	Evidence
Q4	Process to begin to work with interest (must engage with 5%)	1 or	A	e.g. 9000 ÷ 100 × 5 (= 450) **OR** 9000 × 1.05 (= 9450) **OR** 4 × 5 (= 20)
	Develops solution	2 or	AB	e.g. '450' × 4 (= 1800) oe **OR** 9000 ÷ 100 × 5 (= 450) **and** 11800 – 9000 (= 2800)
	Full process to find figures to compare	3 or	ABC	e.g. 9000 + '450' × 4 (= 10800) **or** 9000 + '1800' (= 10800) **OR** '450' × 4 (= 1800) oe **and** 11800 – 9000 (= 2800)
	Valid decision with accurate figure(s)	4	ABCD	e.g. No **AND** 10800 **OR** No **AND** 1800 **and** 2800
	Total marks for question	**4**		

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Section B (Calculator)

PMAT1/C05				
Question	**Process**	**Mark**	**Mark Grid**	**Evidence**
Q1	Begins process to work with money earned or finds total hours worked	1 or	A	e.g. 13.55 × 5 (= 67.75) **OR** 13.55 × 7½ (= 101.625) **OR** 5 + 7h30m + 6h30m (= 19) **OR** 5 + 7.5 + 6.5 (= 19)
	Full process to find total earnings	2 or	AB	e.g. 13.55 × '19' (= 257.45) **OR** '67.75' + '101.625' + '88.075' (= 257.45)
	Accurate figure	3	ABC	257.45
	Total marks for question	**3**		

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Question	Process	Mark	Mark Grid	Evidence
Q2(a)	Begins process to find mean	1 or	A	184 + 150 + 172 + 168 + 150 + 179 + 175 + 182 (= 1360)
	Full process to find mean	2 or	AB	'1360' ÷ 8 (= 170)
	Accurate figure	3	ABC	170
Q2(b)	Valid reverse check	1	D	e.g. 170 × 8 = 1360
	Total marks for question	**4**		

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Question	Process	Mark	Mark Grid	Evidence
Q3	Process to find a missing length	1	A	6 − 2 (= 4) **OR** 10 − 3 (= 7)
	Process to find one relevant area	1 or	B	e.g. 10 × 2 (= 20) **OR** '4' × 3 (= 12) **OR** 10 × 6 (= 60) **OR** '7' × '4' (= 28) **OR** 6 × 3 (=18) **OR** '7' × 2 (= 14)
	Full process to calculate total area	2	BC	e.g. '20' + '12' (= 32) **OR** '60' − '28' (= 32) **OR** '18' + '14' (= 32)
	Process to calculate number of packs	1 or	D	{area}÷ 1.2 (= 26.6..)
	Accurate figure	2	DE	27
	Total marks for question	**5**		

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Question	Process	Mark	Mark Grid	Evidence
Q4	Rounds numbers to manageable figures	1	A	e.g. 100 **or** 105 **or** 25 **or** 30 **or** 26 **or** 27
	Correct use of a multiplication process	1 or	B	e.g. 100 × 25 (= 2500) **OR** 100 × 30 (= 3000) **OR** 100 × 26 (= 2600) **OR** 100 × 27 (= 2700) **OR** 105 × 25 (= 2625) **OR** 105 × 30 (= 3150) **OR** 105 × 26 (= 2730) **OR** 105 × 27 (= 2835) **OR** 103 × 26.99 (= 2779.97)
	Accurate figure using at least one rounded figure from supportive working	2	BC	e.g. 2500 **or** 3000 **or** 2625 **or** 3150 **or** 2730 **or** 2835
	Total marks for question	**3**		

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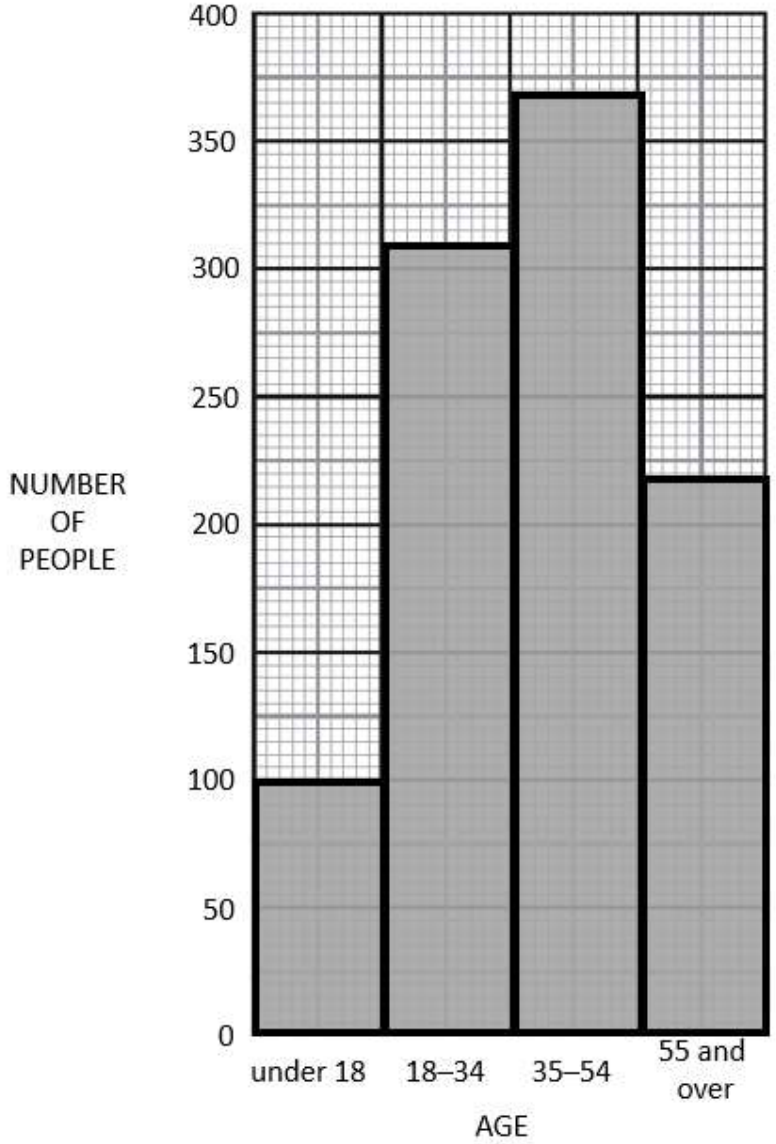
Question	Process	Mark	Mark Grid	Evidence
Q5	Begins to work with angles or proportion	1 or	A	e.g. 360 ÷ (25+15+20) (= 6) **OR** 150 ÷ 25 (= 6) **OR** 15 ÷ (25+15+20) (= 0.25) **or** 20 ÷ (25+15+20) (= 0.33…) May be seen in subsequent calculations
	Process to find one angle or draw 1 angle correctly	2 or	AB	e.g. '6' × 15 (= 90) **or** '6' × 20 (= 120) oe **OR** '0.25' × 360 (= 90) **or** '0.33…' × 360 (= 120) oe May be indicated by one accurately drawn angle
	Fully correct and labelled pie chart	3	ABC	90° (±2°) **and** 120° (±2°) **and** labelled correctly **OVERLAY REQUIRED**
	Total marks for question	**3**		

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Question	Process	Mark	Mark Grid	Evidence
Q6(a)	Begins to draw graph or bar chart	1 or	A	One of: Linear scale Completes labels on horizontal and vertical axes Accurate plotting
	Develops graph or bar chart	2 or	AB	Two of: Linear scale Completes labels on horizontal and vertical axes Accurate plotting
	Fully correct graph or bar chart	3	ABC	All of: Suitable linear scale Completes labels on horizontal and vertical axes Accurate plotting Minimum labels Under 18, 18-34, 35-54, >55, age, people
Q6(b)	Full process to find figures to compare	1 or	D	e.g. 370 ÷ 1000 × 100 (=37) **OR** 370 ÷ 1000 (=0.37) **and** 5 ÷ 100 (=0.05)
	Valid decision with accurate figures	2	DE	e.g. No **AND** 37% **OR** No **AND** 0.37 **and** 0.05 **OR** No **AND** 50
Q6(c)	Completes the frequency column	1	F	4, 12, 16, 8
	Total marks for question	**6**		

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Example Answer for Question 6a Graph



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Question	Process	Mark	Mark Grid	Evidence
Q7	Begins to interpret net	1 or	A	2 of v = 4 w = 8 x = 5 y = 6 **OR** 4, 5, 6, 8 shown but unallocated
	Develops solution	2 or	AB	3 of v = 4 w = 8 x = 5 y = 6
	Accurate figure	3	ABC	All of v = 4 w = 8 x = 5 y = 6
	Total marks for question	**3**		

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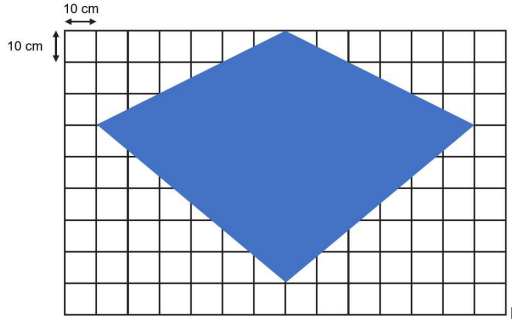
Question	Process	Mark	Mark Grid	Evidence
Q8	Begins to work with fractions	1	A	$1 - \frac{1}{3} (= \frac{2}{3})$ **OR** $24 \div 3$ (=8)
	Full process to find number of people in sales	2 or	AB	$24 \div 3 \times 2$ (=16) **OR** 24 – '8' (=16) oe
	Accurate figure	3	ABC	16
	Total marks for question	**3**		

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Question	Process	Mark	Mark Grid	Evidence
Q9(a)	Begins to work with probability	1 or	A	$\frac{a}{8}$ and $a < 8$ **OR** $\frac{3}{b}$ and $b > 3$
	Accurate figure	2	AB	$\frac{3}{8}$ oe
Q9(b)	Starts to work with bearings	1 or	C	180 + 45 **OR** 360 – 135 oe
	Accurate figure	2	CD	225
	Total marks for question	**4**		

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Question	Process	Mark	Mark Grid	Evidence
Q10	Begin to work with constraints	1 or	A	Draws a kite with 1 of • height 8 squares • width 12 squares • 1 line of symmetry **OR** 120 ÷ 10 (=12) **and** 80 ÷ 10 (=8)
	Develops solution	2 or	AB	Draws a kite with 2 of • height 8 squares • width 12 squares • 1 line of symmetry
	Fully correct diagram	3	ABC	Draws a kite with all of • height 8 squares • width 12 squares • 1 line of symmetry
	Total marks for question	**3**		



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Question	Process	Mark	Mark Grid	Evidence
Q11(a)	Process to use consistent units	1	A	e.g. 0.25 × 100 (= 25) **OR** 0.3 × 100 (= 30) **OR** 0.6 × 100 (= 60) **OR** '0.045' × 100 × 100 × 100 (= 45000) May be seen in subsequent working
	Process to find volume	1	B	e.g. '25' × '30' × '60' (= 45000) **OR** 0.25 × 0.3 × 0.6 (= 0.045)
	Process to convert between litres and cm^3	1 or	C	{volume} ÷ 1000 (= 45) oe **OR** 38.5 × 1000 (=38500)
	Valid decision with accurate figures	2	CD	Mile **AND** 45 (litres) **OR** Mile **AND** 45000 (cm^3) **and** 38500 (cm^3)
Q11(b)	Valid reverse check	1	E	e.g. 45 × 1000 = 45000
	Total marks for question	**5**		